Clinical Trial Analytics Project

Diabetes Treatment Efficacy Analysis

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Project Context

- Clinical Trial Overview
- This project analyzes a clinical trial dataset comparing Auralin and Novodra treatments for diabetes.
- Auralin is a new oral insulin treatment, while Novodra is an existing injectable insulin.
- The primary objective is to analyze the efficacy of Auralin compared to the standard treatment.
- Excel, Power Query, and Power BI were used for end-to-end analytics.

Project Objective

- Objectives of the Project
- Clean and validate raw clinical trial data.
- Transform and merge datasets using Power Query.
- Create Excel and Power BI dashboards.
- Develop DAX measures for KPI reporting.
- Analyze HbA1c reduction and adverse reactions.

Dataset Overview

- Patients Dataset: 503 rows and 17 columns
- Treatments Dataset: 350 rows and 11 columns
- Includes patient demographics, HbA1c values, dosage, BMI, and adverse drug reactions.
- Datasets were cleaned and merged using common identifiers.

Data Cleaning & Validation

- Excel & Power Query Steps
- Removed duplicates and handled missing values.
- Corrected inconsistent date and numeric formats.
- Performed data validation for accurate reporting.
- Changed data types and standardized formats.
- Merged patient and treatment datasets using Power Query.

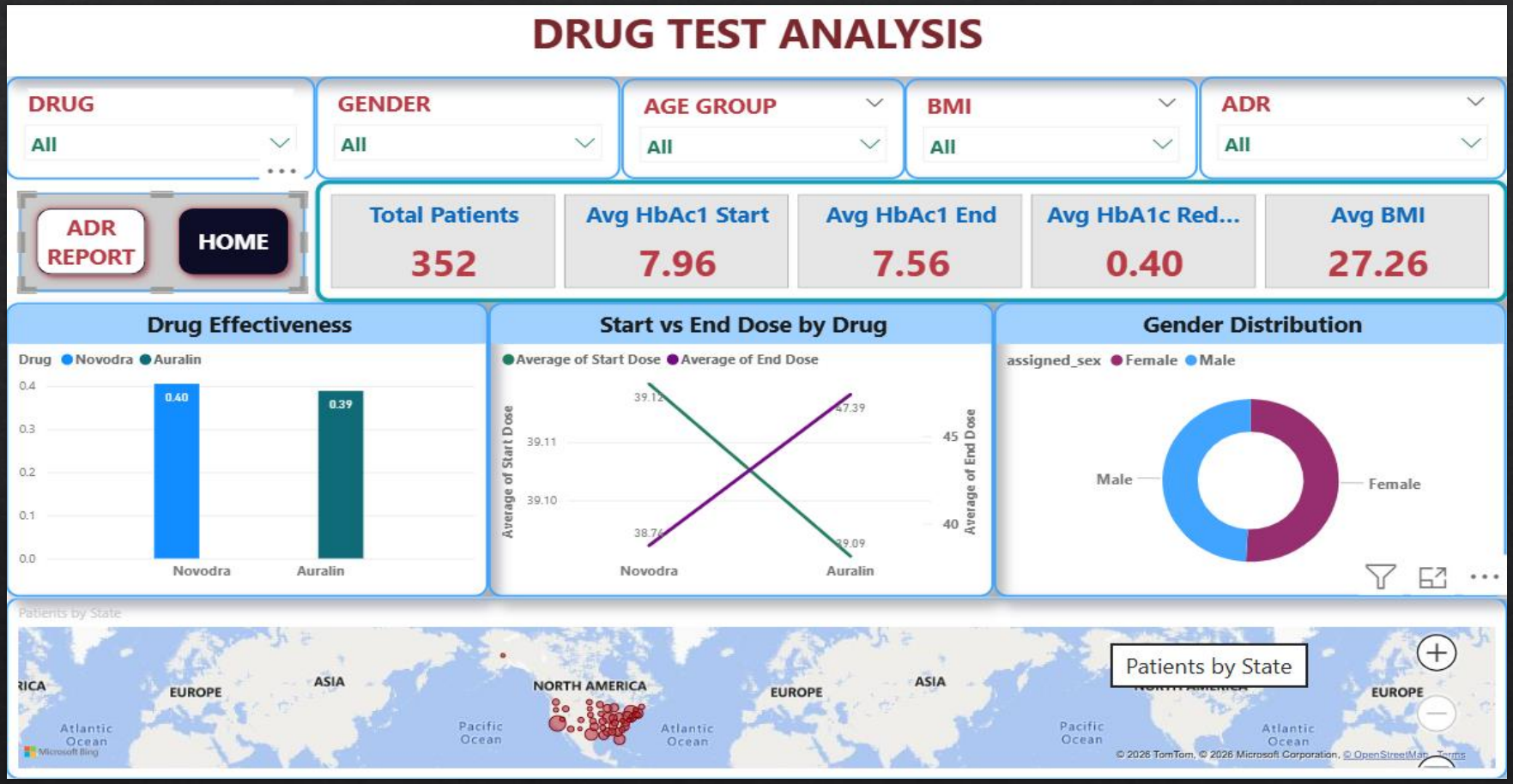
Excel Dashboard



Power BI Dashboard

- Power BI Features
- Imported cleaned data into Power BI.
- Created relationships and DAX measures.
- Developed KPI cards and interactive visualizations.
- Used slicers, filters, and drill-down charts.
- Created ADR analysis and patient trend reports.

Power BI Dashboard



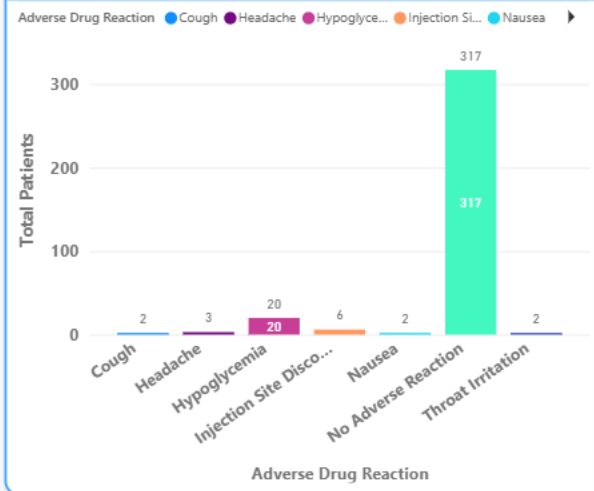
DRUG TEST ANALYSIS

DRUG: All | GENDER: All | AGE GROUP: All | BMI: All | ADR: All

ADR REPORT | HOME

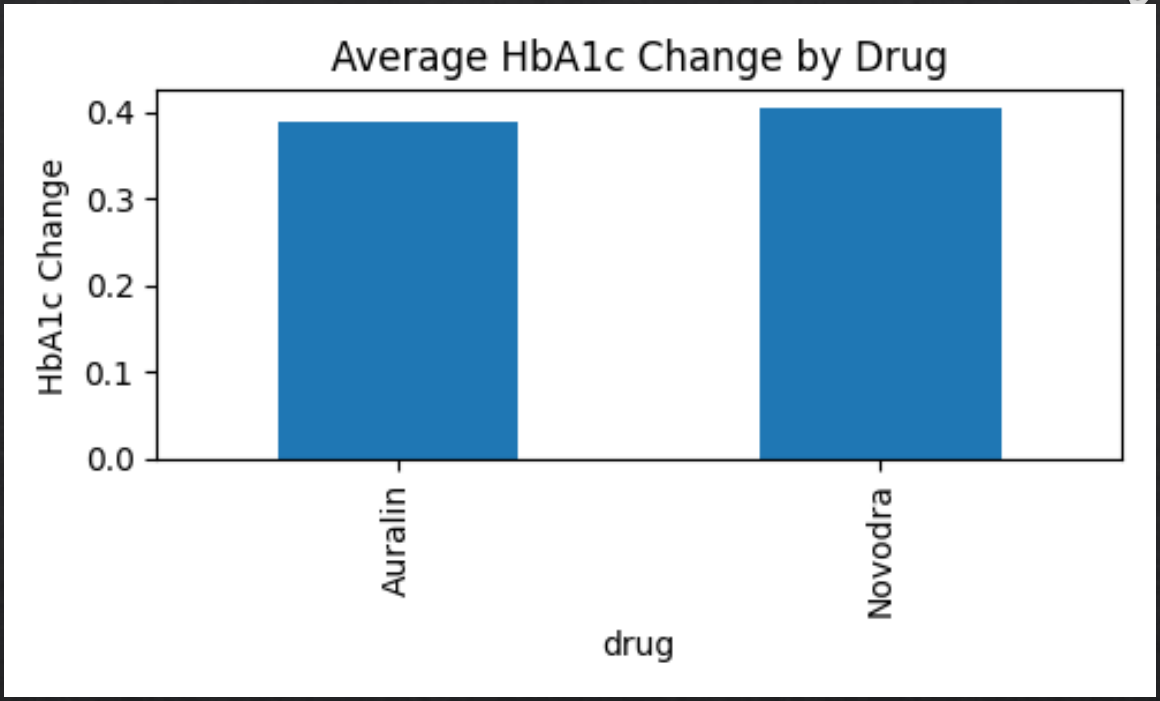
Total Patients	Avg HbAc1 Start	Avg HbAc1 End	Avg HbA1c Red...	Avg BMI
352	7.96	7.56	0.40	27.26

Adverse Reactions by Type VS Patients



BMI chart	Normal		Obese		Over Weight		
Age Group	Count of patient_id	Count of Age	Count of patient_id	Count of Age	Count of patient_id	Count of Age	Count of pa
Middle	42	42	28	28	16	16	
Senior	76	76	88	88	72	72	
Young	13	13	2	2	8	8	
Total	131	131	118	118	96	96	

HbA1c Reduction Analysis



DAX Measures & KPIs

- Key KPI Measures Used
- Total Patients
- Average HbA1c Start
- Average HbA1c End
- Average HbA1c Reduction
- Average BMI
- Drug Effectiveness
- Adverse Drug Reaction Count

Visualizations Used

- Charts and Reports
- KPI Cards
- Bar Charts
- Line Charts
- Donut Charts
- Maps
- Matrix Tables
- Slicers and Filters

Key Insights

- Insights from the analysis:
- Auralin showed measurable HbA1c reduction.
- Comparison with Novodra helps evaluate effectiveness of oral insulin.
- Adverse reaction patterns were monitored for patient safety.
- Dataset is suitable for Power BI dashboard visualization.

Challenges Faced

- Project Challenges
- Handling missing and inconsistent values.
- Correcting incorrect date formats.
- Merging datasets with common identifiers.
- Creating effective dashboard layouts.
- Building accurate DAX measures.

Conclusion

- Project Summary
- Successfully cleaned and transformed clinical trial data using Excel and Power Query.
- Developed interactive dashboards in Excel and Power BI.
- Created DAX measures for KPI and analytical reporting.
- The clinical trial comparison showed that Auralin demonstrated comparable HbA1c reduction effectiveness against Novodra.
- The analysis also indicated that most patients experienced no major adverse drug reactions.
- The project demonstrated practical healthcare analytics and data visualization workflow.

Thank You

- Questions & Discussion